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CONTRIBUTION

Source: Ministry of Communications, India; Telecom Regulatory Authority of India, India; Telecommunications Standards Development Society, India; Centre for Development of Telematics, India

Title: Proposed baseline text for the proposed draft new ITU-T Recommendation J.FRCAS "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content".

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Abstract: This contribution contains the proposed baseline text for a proposed draft new ITU-T Recommendation on "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content".

Summary and Rationale: The proposed Recommendation will specify the functional requirements of Conditional Access Systems (CAS) deployed in Digital Addressable Systems (DAS) for television broadcasting to enable factual subscriber-base reporting and to ensure protected delivery of content to authorised subscribers.

The objective of the proposed Recommendation is to address two major concerns related to CAS, namely, underreporting of subscribers and content piracy.

This draft new Recommendation will elaborate the various functional requirements of the CAS such as log requirements, reports requirements, database requirements, security requirements, service requirements, and more. Compliance to these requirements in CAS performance will address the two concerns mentioned above.

Proposal: The draft baseline text for the proposed draft new ITU-T Recommendation J.FRCAS is placed as ANNEX-1.

Recommendation ITU-T J.FRCAS

Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content

Summary

Most countries have implemented Digital Addressable Systems (DAS) for distribution platforms such as Cable TV, Direct to Home (DTH), Internet Protocol Television (IPTV), and Headend In The Sky (HITS). DAS enables subscribers to have a choice and option to select their channels and packages. Conditional Access System (CAS) and Subscriber Management System (SMS) are integral parts of a DAS environment. While CAS is responsible for encrypting the content and its delivery in a secure and encrypted manner only to authorised subscribers, SMS handles the subscriber management functions. The quality of service depends on the CAS and SMS systems deployed by the Distribution Platform Owner (DPO).

Since the content providers and broadcasters have no control over the CAS and SMS, as these are deployed at the DPO end, they often complain about the loss of revenue due to various malpractices, such as under-reporting of subscriber-base, distribution of channels to unauthorised users, and content piracy.

Therefore, it is necessary to frame benchmarks for the CAS systems, prescribing technical specifications focusing on functional requirements to ensure factual reporting of subscriber-base and protected delivery of content to authorised subscribers. Creating a framework that prevents the deployment of sub-standard CAS will bring preventive control from potential threats and revenue losses arising due to the vulnerability of such systems, bring economic efficiency, improve quality of service, and improve the end-consumer experience.

The objective of the Recommendation is to address two major concerns related to CAS, namely, underreporting of subscriber numbers and content piracy, leading to revenue loss to broadcasters, content providers and the governments.

This Recommendation elaborates the various functional requirements of the CAS such as log requirements, reports requirements, database requirements, security requirements, service requirements, and more. Compliance to these requirements in CAS performance will address the concerns mentioned above.

Keywords

Conditional Access, Subscriber Management System, CAS, SMS, piracy, misreporting

Change Log

This document contains Version 1.0 of the draft ITU-T Recommendation J.FRCAS on "Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content" approved at the ITU-T Study Group 09 meeting held in Geneva, 06-14 Sept. 2022.

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Recommendation ITU-T J.FRCAS

Functional requirements of Conditional Access Systems (CAS) for factual subscriber-base reporting and protected delivery of content

1 Scope

The Recommendation specifies the functional requirements of Conditional Access Systems (CAS) deployed in Digital Addressable Systems (DAS) for television broadcasting to enable factual subscriber-base reporting and to ensure protected delivery of content to authorised subscribers.

The objective of the Recommendation is to address two major concerns related to CAS, namely, underreporting of subscriber numbers and content piracy, leading to revenue loss to broadcasters, content providers and the governments.

This Recommendation elaborates the various functional requirements of the CAS such as log requirements, reports requirements, database requirements, security requirements, service requirements, and more. Compliance to these requirements in CAS performance will address the concerns mentioned above.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

- [1] [ITU-R BT.1852-1] Recommendation ITU-R BT.1852-1 (2016), *Conditional-access systems for Digital broadcasting*.
- [2] [ITU-T J.95] Recommendation ITU-T J.95 (1999), *Copy protection of intellectual property for Content delivered on cable television systems*.
- [3] [ITU-T J.93] Recommendation ITU-T J.93 (1998), *Requirements for conditional access in the Secondary distribution of digital television on cable television systems*.

3 Terms and definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 Conditional Access system (CA) [ITU-T J.95]: The complete system for ensuring that cable services are accessible only to those who are entitled to receive them, and that the ordering of such services is not subject to modification or repudiation.

3.1.2 Encryption [ITU-T J.93]: The process of scrambling signals to avoid unauthorized access.

3.1.3 Piracy [ITU-T J.93]: The act of acquiring unauthorized access to programs, usually for the purpose of reselling such access

3.1.4 Scrambling [ITU-T J.93]: The process of using an encryption function to render television and data signals unusable to

3.1.5 Content [ITU-R BT.1852-1]: This is any form of digital data that can be acquired and presented by a device.

3.1.6 **Service** [ITU-R BT.1852-1]: This is one or more data flows intended to be presented together.

3.2 Terms defined here

This Technical Report defines the following terms:

3.1.1 **term** [**reference**]: **definition**

3.1.2 **term** [**reference**]: **definition**

[To be added as the study progresses.]

4 Abbreviations

API	Application Programming Interface
CAS	Conditional Access System
DB	Database
DPO	Distribution Platform Operator
DTH	Direct To Home
EPG	Electronic Programme Guide
FP	Finger Printing
FTA	Free-To-Air
HITS	Headend In The Sky
IPTV	Internet Protocol Television
LCN	Logical Channel Number
OTA	Over-The-Air
OTT	Over-The-Top
PIN	Postal Index Number
SLA	Service-Level Arrangement
SMS	Subscriber Management System
STB	Set Top Box
TEC	Telecommunication Engineering Centre
TRAI	Telecom Regulatory Authority of India
UA	Unique Access
UI	User Interface
VC	Viewing Card

5 Conventions

None

6 Overview

Most countries have implemented Digital Addressable Systems (DAS) for distribution platforms such as Cable TV, Direct to Home (DTH), Internet Protocol Television (IPTV), and Headend In The Sky (HITS). DAS enables subscribers to have a choice and option to select their channels and packages. Conditional Access System (CAS) and Subscriber Management System (SMS) are

integral parts of a DAS environment. While CAS is responsible for encrypting the content and its delivery in a secure and encrypted manner only to authorised subscribers, SMS handles the subscriber management functions. The quality of service depends on the CAS and SMS systems deployed by the Distribution Platform Owner (DPO).

Since the content providers and broadcasters have no control over the CAS and SMS, as these are deployed at the DPO end, they often complain about the loss of revenue due to various malpractices, such as under-reporting of subscriber-base, distribution of channels to unauthorised users, and content piracy.

Several countries have enabling provisions for mandatory audit of the DPO systems to provide requisite assurance to broadcasters and content providers, and CAS and SMS systems have to conform to certain technical features to check piracy and other malpractices.

Multiple CAS solutions are available, some using advanced embedded security while others using sub-standard security solutions. Systems having sub-standard solutions can be prone to manipulations and vulnerable to hacking, leading to loss of revenue to the concerned broadcasters and the government.

Therefore, it is necessary to frame benchmarks for the CAS systems, prescribing technical specifications focusing on functional requirements to ensure factual reporting of subscriber-base and protected delivery of content to authorised subscribers. Creating a framework that prevents the deployment of sub-standard CAS will bring preventive control from potential threats and revenue losses arising due to the vulnerability of such systems, bring economic efficiency, improve quality of service, and improve the end-consumer experience.

7 Functional Requirements

The CAS should meet the following functional requirements:

7.1 Log Requirements

7.1.1 CAS Logs

7.1.1.1. CAS logs such as the user command, configuration, channel/ bouquet creation, modification, etc., shall be kept in a secured and un-editable way.

7.1.2 Time Stamping

7.1.2.1. All logs shall be stamped with date and time.

7.1.2.2. The system shall not allow altering or modification of any logs. There shall be no facility for the distributor/ users to purge logs.

7.1.3 SMS and CAS Integration

7.1.3.1 Each instance of the activity carried out at SMS pertaining to CAS shall be recorded in the logs/ reports of CAS, along with date and time stamp.

7.1.3.2. If any activity has been carried out directly from CAS for troubleshooting; such an exception shall be identified through the synchronization mismatch report.

7.1.3.3. For any activity outside the normal channel/ route of SMS-based commands, a secure log shall be maintained and made available on request to the audit or testing agency for scrutiny.

7.2 Report Requirements

7.2.1. CAS database (DB) shall have the reports of white list of card/ STBs along with details such as active/ inactive status, with the date and time stamp.

- 7.2.2. CAS system shall be capable of generating reports pertaining to the channel/ bouquet subscriptions and active/ deactivated subscribers, or any combination thereof; of sharing the same with SMS as a scheduled activity, and also upon request, including, but not limited to, the following details:
 - 7.2.2.1. STB Number
 - 7.2.2.2. Viewing Card (VC) Number [or, in case of card-less CAS, chip identification (ID) or virtual card number of the STB]
 - 7.2.2.3. Product Code pertaining to channels/ bouquets available on the platform
 - 7.2.2.4. Start date of entitlement
 - 7.2.2.5. End date of entitlement
 - 7.2.2.6. Status of card (Active/Inactive)
- 7.2.3. It shall be possible to generate following reports from the logs of CAS:
 - 7.2.3.1. STB-VC pairing/ de-pairing
 - 7.2.3.2. STB activation/ deactivation
 - 7.2.3.3. Channel assignment to STB
 - 7.2.3.4. Report of the activation/deactivations of a particular channel for a given period

7.3 CAS Database Requirements

- 7.3.1. There shall not be any active unique subscriber outside the database tables.
- 7.3.2. There shall not be an option to split CAS database for creation of more than one instance by a DPO or a vendor.
- 7.3.3. CAS must support the following options with reference to uploading of unique access (UA)/ viewing card (VC) details in CAS database:
 - 7.3.3.1. A secure un-editable file of card details, as purchased by the distributor, to be uploaded by the CAS vendor on the CAS Server directly, or,
 - 7.3.3.2. If it is uploaded in any other form, UA/ VC in CAS database shall be captured in logs.
 - 7.3.3.3. CAS shall support an automated, application programming interface (API)-based mechanism to populate such UA/ VC details in the SMS, without any manual intervention.
- 7.3.4. CAS shall ensure that the access to database is available to authorized users only, and in “read only” mode only.
- 7.3.5. The database audit trail shall be permanently enabled. [Explanation: Database here refers to the database where data and log of all activities related to STB activation, deactivation, subscription data, STB UA/VC details, entitlement level information, etc., is being stored.]
- 7.3.6. CAS shall have a provision to export the database/ report for reconciliation with the SMS database.
- 7.3.7. There shall be a provision of reconciliation through secure APIs/secure scripts.
- 7.3.8. CAS and SMS Server Separation: CAS and SMS applications, along with their respective databases, shall be stored in such a way that they can be separately identified.
- 7.3.9. CAS Backup Server
 - 7.3.9.1. In the event of provisioning of a backup server, logs of all activities carried out in main server shall be concurrently copied into the backup server:

- 7.3.9.2. Provided that a log of all such instances shall be maintained along with date and time stamp, where the backup server has been used as the main server:
- 7.3.9.3. Provided further that the main and backup server shall always be in sync with regard to the key data such as subscription data, STB UA/ VC details, entitlement level information, etc.

7.4 Security Requirements

7.4.1 Activation and Deactivation

- 7.4.1.1. No access/login IDs/user interface/application shall be provided to the distributor of television channels to execute any commands, including but not limited to, activation/deactivation, bouquet creation/modification/deletion, etc., directly from CAS by bypassing SMS.

7.4.2 Logical Channel Number (LCN)

- 7.4.2.1. CAS shall not support carriage of channel with same name or nomenclature in the distributor's network served by each headend under more than one LCN, and another channel descriptor.
- 7.4.2.2. Each channel available in CAS shall be uniquely mapped with channels available in SMS.
- 7.4.3. CAS shall be accessible through a Firewall only.

7.4.3 CAS Server Hardware

- 7.4.4.1. CAS shall be deployed on hardened secure server hardware.
- 7.4.4.2. CAS shall protect against any backdoors, malicious software deployments, and cyber security threats.

7.4.4 Finger printing (FP) measures

- 7.4.5.1. CAS shall support both covert and visible types of finger printing functionality.
- 7.4.5.2. The fingerprinting shall be on the topmost layer of the video.
- 7.4.5.3. The fingerprinting shall appear on the screen in all scenarios, such as menu, electronic programme guide (EPG), settings, blank screen, games, etc.
- 7.4.5.4. The fingerprinting shall not get invalidated by use of any device or software.
- 7.4.5.5. CAS shall have the capability to run fingerprinting at regular intervals (e.g., minimum of 2 fingerprints per hour on a 24x7x365 basis) and provide broadcasters with the fingerprint schedule on request.
- 7.4.5.6. The fingerprinting shall be available on global as well as on individual STB basis.

7.5 Service Requirements

7.5.1 Provision of à-la-carte channels or bouquet

- 7.5.1.1. CAS shall be able to handle all the channels, made available on a platform, in à la carte mode.
- 7.5.1.2. CAS shall have the capability to handle such number of broadcaster/DPO bouquets, as required by the DPO.

7.5.2 Set Top Box (STB) Operation

- 7.5.2.1. Upon deactivation of any subscriber from the SMS, all program/ services, including all free- to-air (FTA) and pay channels and platform services, shall be denied to that subscriber:
- 7.5.2.2. Provided that there shall be a facility for the distribution platform operator (DPO) to continue to provide B-mail/ scroll messages that enable a consumer to get the information in relation to the recharge/ payment of the pending dues.

7.5.3 Channel Addition

- 7.5.3.1. CAS shall be capable to add/ modify channels/bouquets as may be required from time to time.

7.5.4 Hybrid STB

- 7.5.4.2. In case a distributor of television channels has deployed hybrid STBs, CAS shall ensure that the over-the-top (OTT) App does not get access to the linear Television channels, and the CAS does not get access to channels delivered through OTT platform: [Provided that, all the mandatory requirements for CAS shall be complied by the hybrid STBs.]

7.5.5 CAS-STB addressability

- 7.5.5.1. CAS shall be capable of providing STB/viewing card information with the current date, time, and name/logo of the distributor of television channels.
- 7.5.5.2. CAS shall be capable of individually addressing subscribers, for the purpose of generating the reports, on channel by channel and STB by STB basis.
- 7.5.5.3. CAS shall be capable of tagging and blacklisting VC numbers and STB numbers that are involved in piracy, to ensure that such STB/ VC cannot be redeployed.
- 7.5.5.4. CAS shall be capable of upgrading STBs over-the-air (OTA), so that the connected STBs can be upgraded.

7.5.6 De-entitlement of STB

- 7.5.6.1. The entitlement end date in CAS shall be equal to the entitlement end date in SMS, or,
- 7.5.6.2. The entitlement end date in CAS shall be open and SMS shall manage entitlements based on the billing cycles and payments.

7.5.7 Message Queue

- 7.5.7.1. In the event of unsuccessful transmission of messages due to network failure (for instance, due to power failure), the head-end should have an option to queue up the messages.
- 7.5.7.2. There should be a provision to retry them at specified intervals using additive back off retrieval timings.
- 7.5.7.3. In the event of unsuccessful deliveries of the messages, the life of the messages should be specifiable.

7.5.8 Geographical Blackout

- 7.5.8.1. CAS shall have the feature of geographical blackout.[Explanation: Geographical blackout is the ability of CAS to blackout a particular region based on the postal index number (PIN) Codes [Geographic Area Code] if required by government agencies or for other reasons.]

7.5.9 After-Sales Service Support

- 7.5.9.1. The required software and hardware support should be available to the distributor of the television channels' installations from the CAS vendor's support teams located in their respective countries. [Explanation: The requirement for hardware support should be applicable, only if the hardware is directly or indirectly provided by the CAS vendor.]
- 7.5.9.2. The support should be such as to ensure the CAS system with 99.99% uptime and availability. [Explanation: The actual service-level arrangement for the system support shall be governed by the mutual agreement/ service-level agreement (SLA) between the service provider, i.e. CAS vendor and the customer (DPO). The signatories to the said agreement may mutually choose lenient/ stringent service-level guarantee.]
- 7.5.9.3. The systems should have sufficient provisions for backup systems to ensure quality of service and uptime.

8 Bibliography

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- [2] [TEC 57015:2022] Telecommunication Engineering Centre (TEC), Ministry of Communications, India; *Test Guide for Conditional Access System (CAS)*